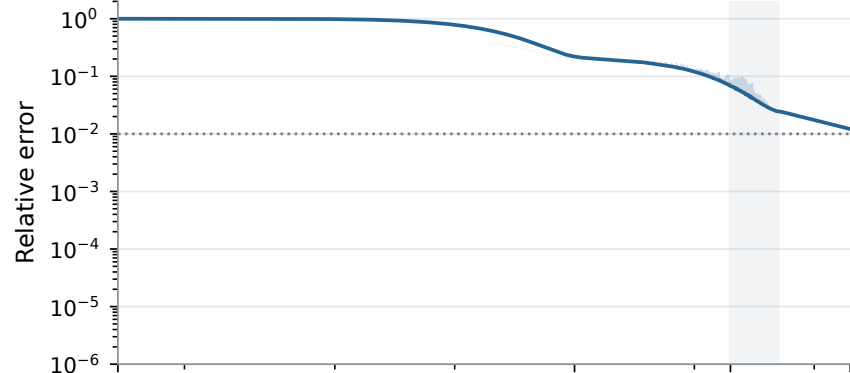


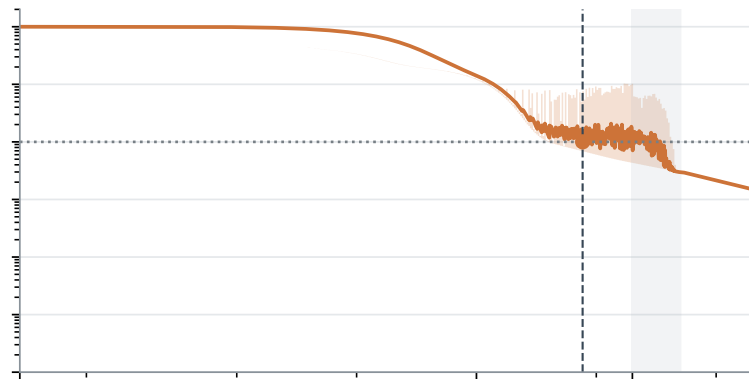
First EMA crossings resolve early acquisition

Sine mixture · common Adam setting · shaded vertical window: learning-rate decay (20k-50k)

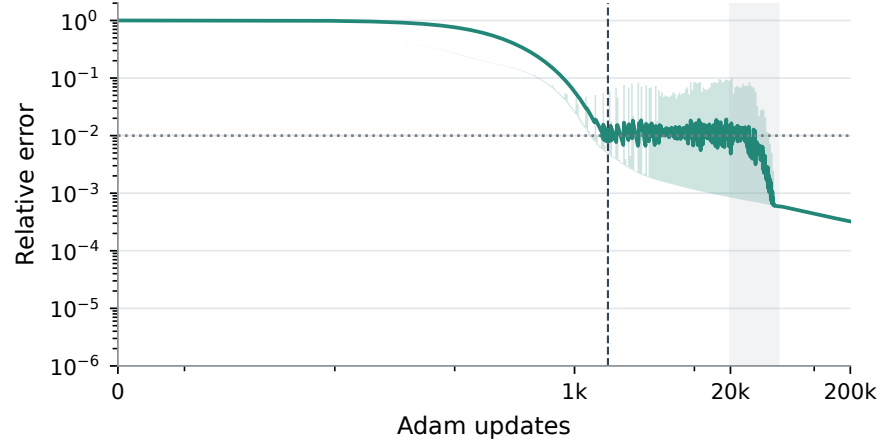
$\gamma = 8$ First EMA crossing: >200,000



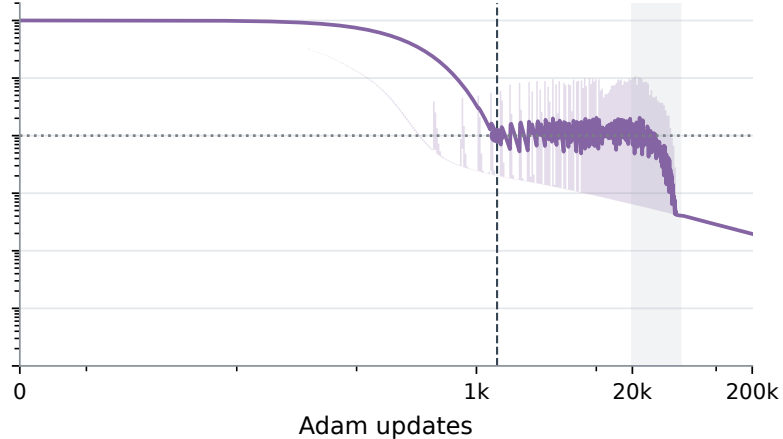
$\gamma = 12$ First EMA crossing: 7,688



$\gamma = 16$ First EMA crossing: 1,900



$\gamma = 64$ First EMA crossing: 1,486



All-update min-max envelope

Square root of loss EMA (half-life 100)